

**CESifo
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**9686
2022**
April 2022

**Visual Representation and
Stereotypes in News Media**

Elliott Ash, Ruben Durante, Maria Grebenshchikova, Carlo Schwarz

VISUAL
REPRESENTATION
AND
STEREOTYPES
IN
NEWS MEDIA

AGENDA

Research Question	3
Previous Work & Limitations	4
Novel Approach	6
Data	8
Main Results	9
Importance	1
Discussion & Questions	9
	2
	0

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RESEARCH QUESTION

How do news images in the *New York Times* (**NYT**) and *Fox News* (**Fox**) from 2000 to 2020 represent...

- gender and ethnic groups, and
- to what extent do these visual representations reinforce common stereotypes or
- reflect the identities of journalists and editors?

PREVIOUS WORK & LIMITATIONS

- Text / Images are only one dimension of news content
- Text-based methods are effective for studying gender due to the presence of gendered nouns and pronouns. But they are less suited to racial or ethnic stereotypes, which are rarely explicit in text

NOVEL APPROACH

Analyzing **images and their relationship to text**

by combining

Computer Vision and **Natural Language Processing** (NLP)

NOVEL APPROACH

Use of Computer Vision:

1. automatically detect faces (91% accuracy)
 - a. classifies gender (91% accuracy)
 - a. classifies ethnicity (73% accuracy)

Use of Natural Language Processing:

standard NLP

DATA

Data Sources: online articles from web pages of NYT and Fox

Data: 2,219,267 articles published from 2000 to 2020

Dataset: all articles that consist of text and at least one image

MAIN RESULTS

1.

Over- / Under-
Representation of
Gender and
Race

2.

Stereotypical
Associations
between Text
and Image

3.

Influence of
News Producers
on Group
Representation

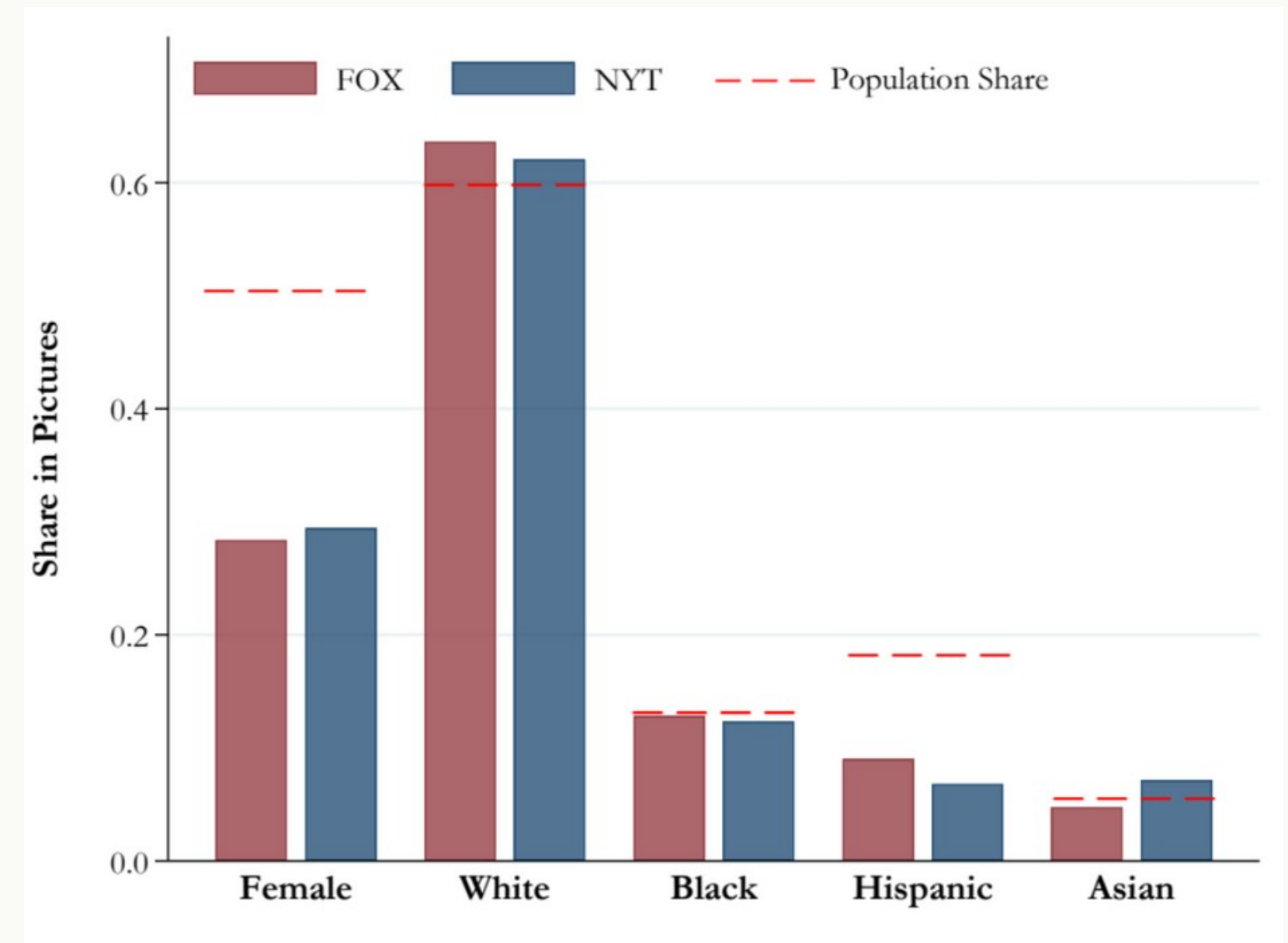
1 Over- / Under-Representation of Gender and Race

Research Question: How often do people from a given identity group show up in articles of Fox and NYT compared to the group's share in the U.S. population overall from 2000 to 2020.

1 Over- / Under-Representation of Gender and Race

Findings: absolute numbers

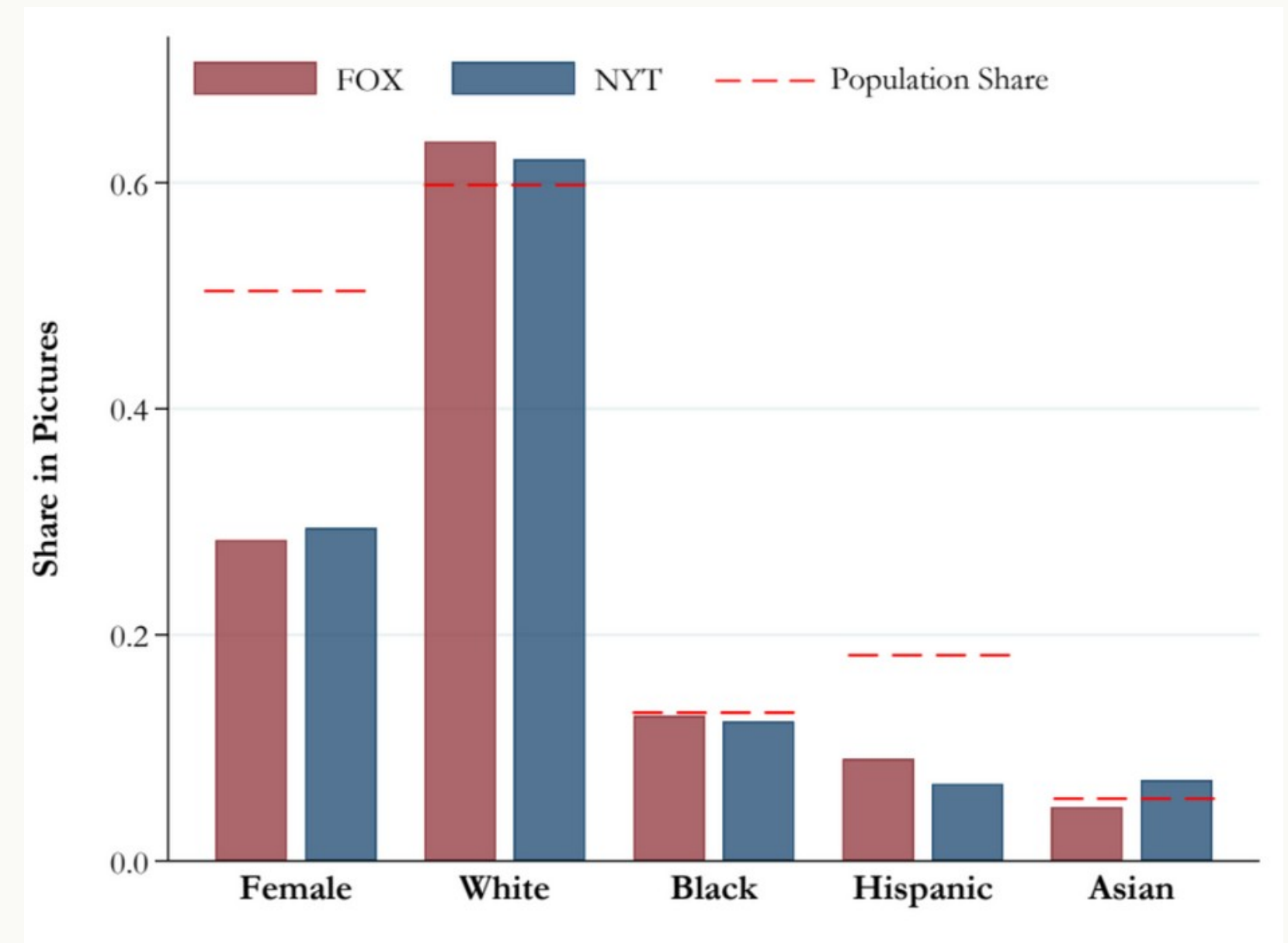
1. Women are heavily under-represented
2. Whites are generally over-represented
3. Blacks and Asians are represented more or less proportionally
4. Hispanics are under-represented



1 Over- / Under-Representation of Gender and Race

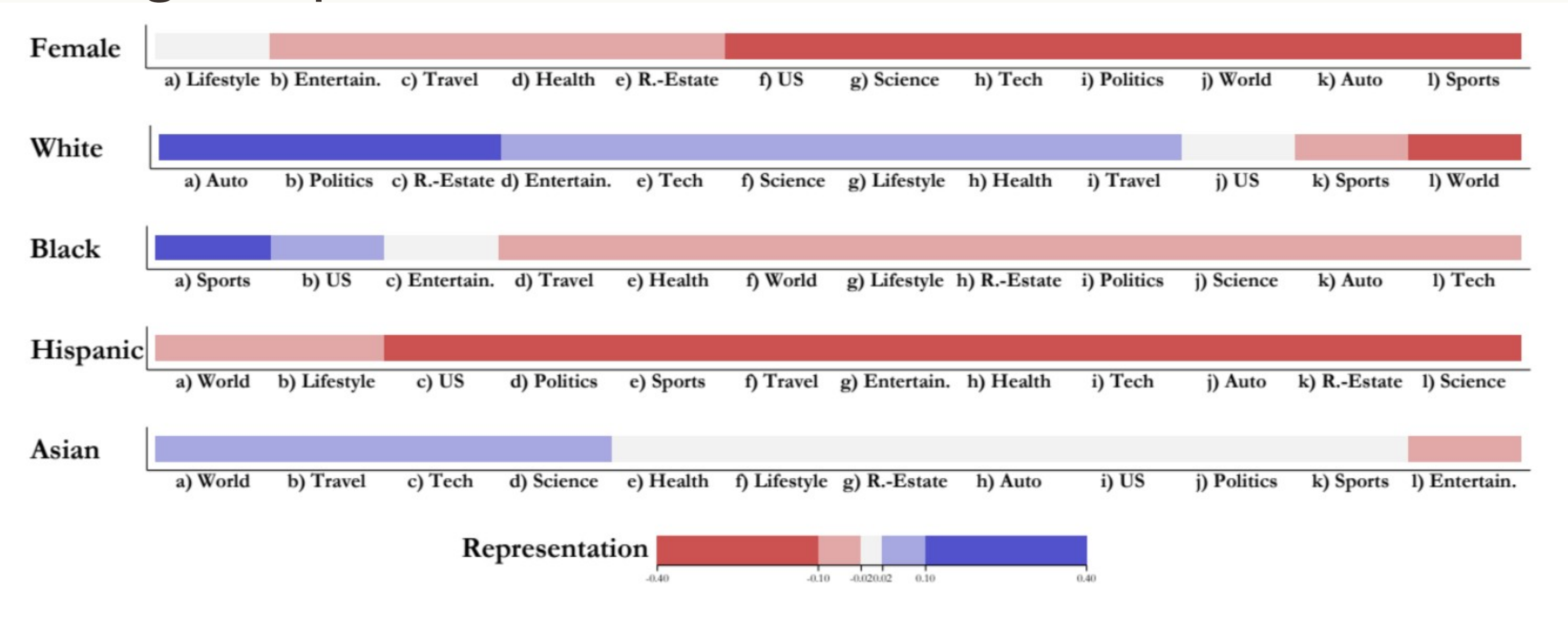
Findings: NYT vs. Fox

1. Compared to NYT, Fox shows fewer images of women and more of Whites
2. Both, NYT and Fox depict Black individuals at comparable rates
3. Compared to NYT, Fox shows more images of Hispanics and fewer images of Asians



1 Over- / Under-Representation of Gender and Race

Findings: representation across sections



1 Over- / Under-Representation of Gender and Race

Findings: representation across sections

1. Women have higher representation in “soft news”, but are otherwise under-represented
2. Whites are generally over-represented
3. Blacks are under-represented in all sections except the Sports section
4. Hispanics and Asians are most associated with the world section, contrary to Blacks
5. Asians are over-represented in tech and science

2 Stereotypical Associations between Text and Image

Research Question: Are certain identity groups of the U.S. population disproportionately linked to particular topics (like crime, poverty, or immigration) or jobs in news articles of Fox and NYT from 2000 to 2020?

2 Stereotypical Associations between Text and Image

Findings:

1. Articles about crime and poverty are disproportionately associated with images of Blacks
2. Articles about immigration are disproportionately associated with images of Hispanics
3. News images display racial stereotypes, overrepresenting groups in articles about stereotypical jobs

3 Influence of News Producers on Group Representation

Research Question: Does the *identity of journalists and editors* of Fox and NYT influence which identity groups of the U.S. population appear in article images published between 2000 and 2020?

3 Influence of News Producers on Group Representation

Findings:

1. Author identity significantly relates to the gender and ethnicity featured in images
2. A more diverse editor team results in more diverse news

IMPORTANCE

- Mass media play an essential role in popularizing gender and ethnic stereotypes
- Linking image to text:
 - Images make up a large fraction of the news content and thus play a central role in newspaper narratives
 - Images have a stronger persuasive effect than text

DISCUSSION & QUESTIONS

1.

Limitations of this Study:

What do you think are the limitations of this study? Do you see any issues with the dataset?

2.

Impact of Representation:

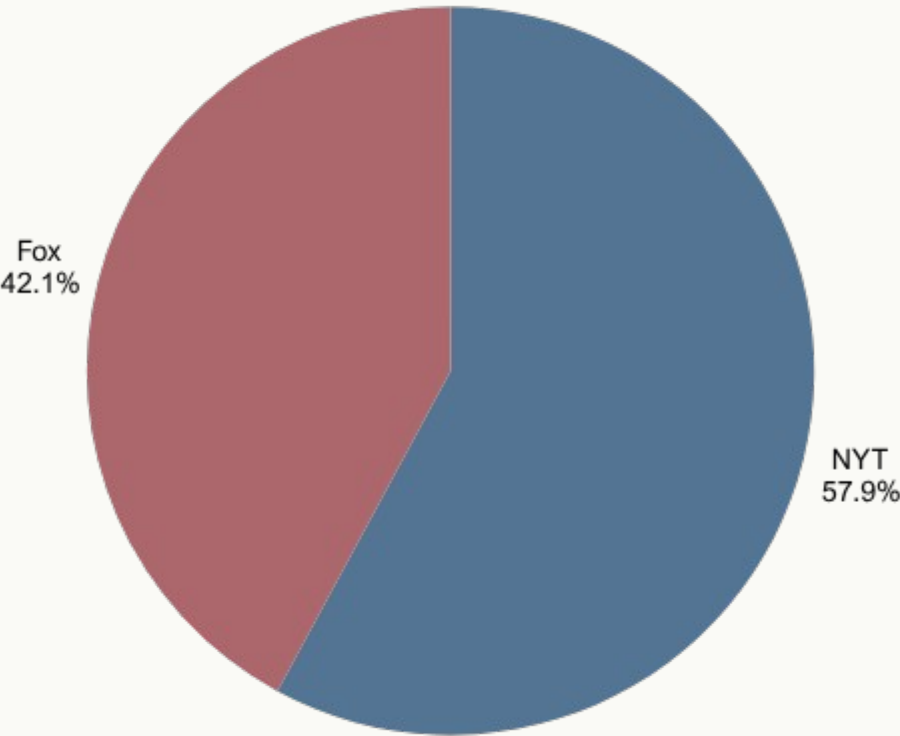
If news outlets systematically over-represent some groups (e.g., White men) and under-represent others (e.g. women, Hispanics), what kinds of real-world social or political effects might this have on the audience?

DATA

Data Sources: online articles from web pages of NYT and Fox
Dataset: 2,219,267 articles published from 2000 to 2020
Sample: all articles that consist of text and at least one image

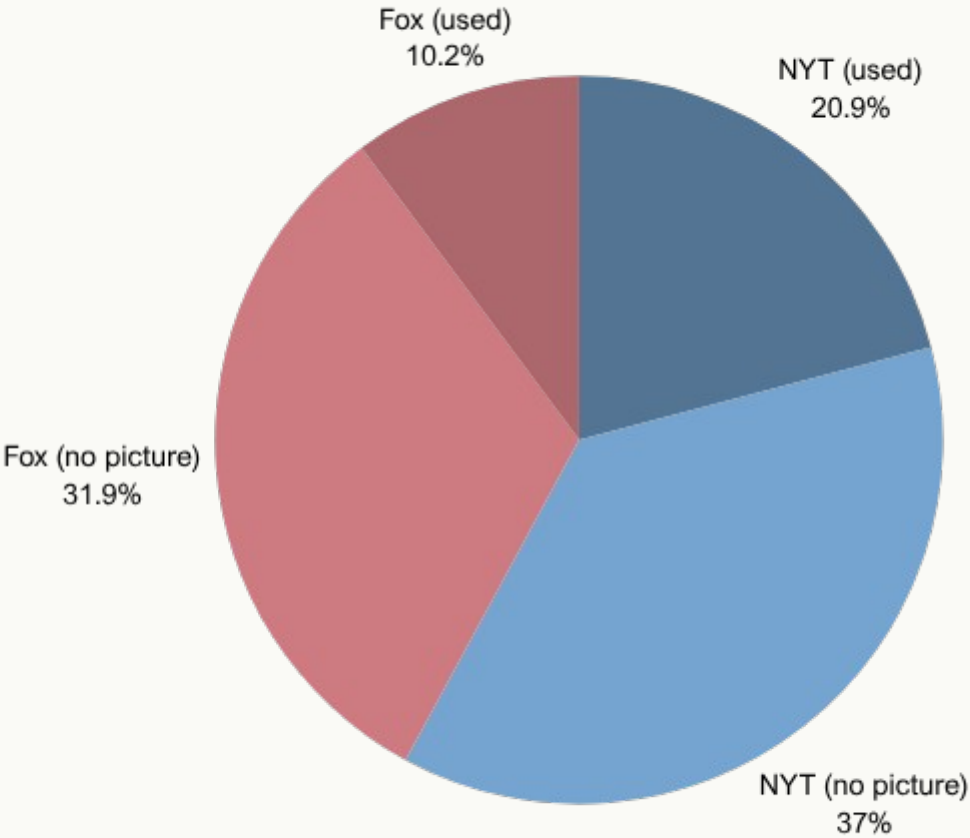
DATA

Total: 2,219,267 (100%)
NYT: 1,284,939 (58%)
Fox: 934,328 (42%)



DATA

Total: 690,152 (31%)
NYT: 463,886 (21%)
Fox: 226,266 (10%)



DATA

Total: 414,091 (19%)
NYT: 278,331 (13%)
Fox: 135,760 (6%)

